basis_label

Gross demand

Net demand (minus wind and solar)

d_h

1

4

2

label

- [x] 2021
- [x] 2022
- [x] 2023
- [x] 2024
- [x] 2025
- [ ] 2026 (check)

window

all

PM

AM

sensitivity

base

usable70

top50

MW, lowest single winter (2021)

p* (lowest winter, 2 h)

234 MW

Break-even fleet size (mean)

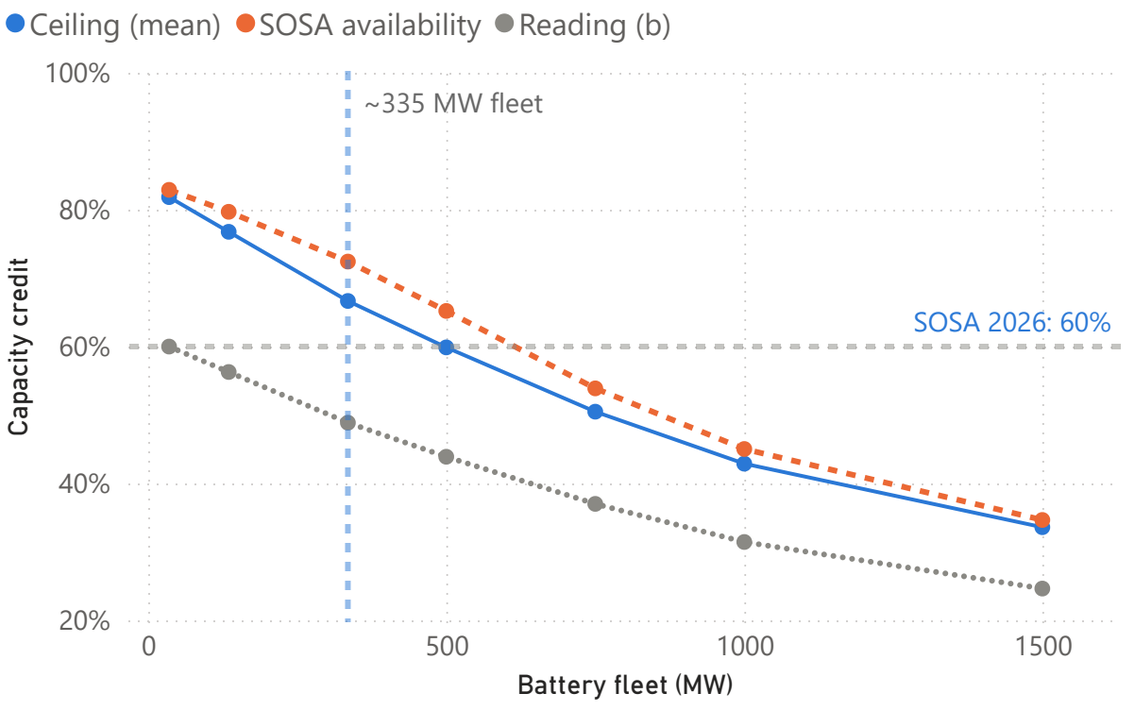
MW where the 2021–2025 mean ceiling falls to 60%

p* (base mean, 2 h)

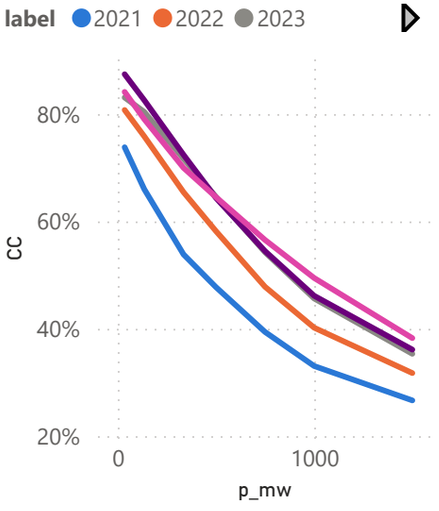
497 MW

Ceiling = what a battery fleet with perfect foresight could lower H100 by, per MW. Real batteries can only do less. Source: EMI public data,

A two-hour battery counts for less as the fleet grows



Each winter on its own



ceiling at today's fleet size

CC (base mean)

66.7%

Source: EMI public data · run 20261019T032856Z_d29a686

Date

21/07/2025 21/07/2025

label

35 MW	500 MW	1,500 MW
135 MW	750 MW	
335 MW	1,000 MW	

basis_label

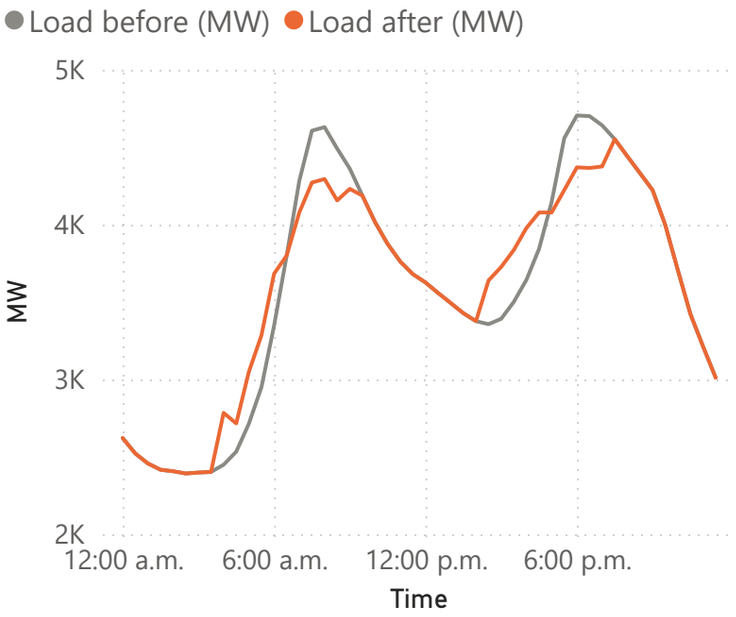
- Gross demand
- Net demand (minus wind and solar)

Peak lowered on this day

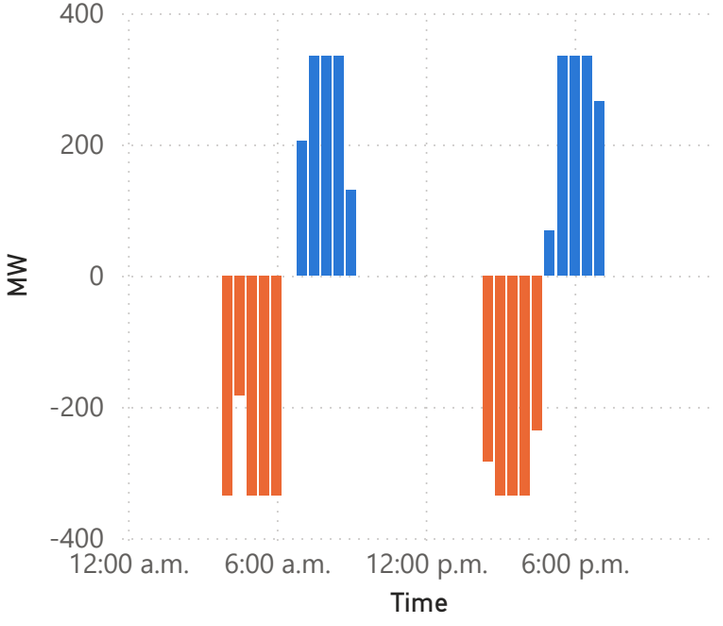
Peak shaved (MW)

153

Load before and after the battery

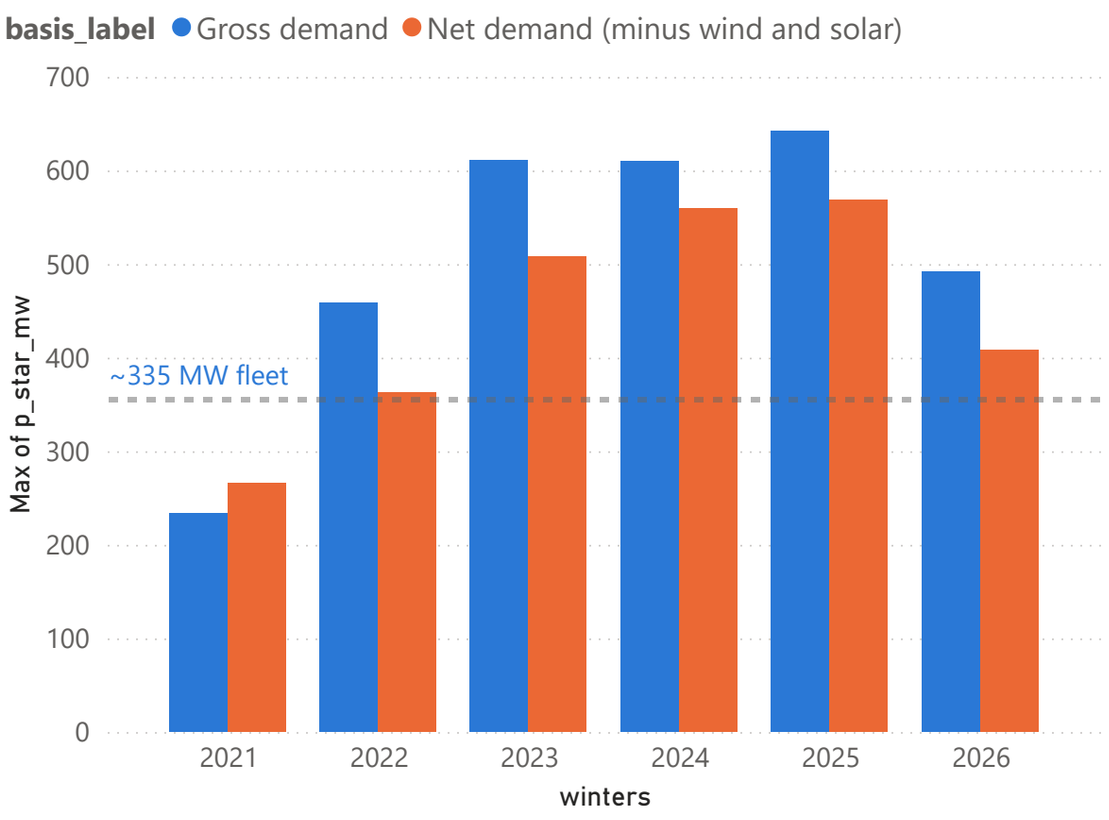


Battery: discharge (+) and charge (-)



Source: EMI public data · run 20261019T032856Z_d29a686

Where each winter's ceiling falls to 60%



winter	units_counted	nameplate_mw	observed_factor	ceiling_availability
2025	Ruakākā	100	0.02	0.82
2026	Ruakākā+Rotohiko	135	0.19	0.75

Energy only:
reserves (FIR/SIR) are not in Generation_MD.

Top 200 on gross vs net demand

winter	basis	overlap	h100_mw	wind_cf	share_pm
2021	gross	64.0%	4,433.78	27.8%	66.5%
2021	net	64.0%	4,235.62	15.5%	69.0%
2022	gross	60.0%	4,399.29	33.2%	76.5%
2022	net	60.0%	4,190.92	13.3%	65.0%
2023	gross	69.0%	4,454.16	28.2%	66.5%
2023	net	69.0%	4,261.43	14.8%	67.0%
2024	gross	55.5%	4,424.12	32.4%	78.0%
2024	net	55.5%	4,186.97	12.9%	67.5%
2025	gross	67.5%	4,456.67	29.3%	64.5%
2025	net	67.5%	4,198.92	16.2%	63.5%
2026	gross	74.0%	4,554.43	26.8%	66.0%
2026	net	74.0%	4,314.94	15.9%	69.0%

Calibration against SOSA 2026 (G4)

year	tech	cf	sosa_cf	diff_pp
2024	solar	0.05	0.01	3.49
2024	wind	0.33	0.31	1.63
2025	solar	0.06	0.07	-1.16
2025	wind	0.28	0.27	1.25

Source: EMI public data · run 20261019T032856Z_d29a686